Emergency Department Arrival Study — Executive Summary

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I. The Folk Wisdom Nobody Has Formally Tested

Go by ambulance and you get seen faster. It is one of the most widely repeated pieces of advice about navigating an ER. It is also something hospitals and EMS agencies actively push back on. Their official position — published on hospital websites and repeated in public health communications — is that ambulance arrival does not get you seen faster. Triage is the system. Clinical severity determines wait time. How you arrived does not matter.

This audit tested that claim against a nationally representative federal dataset of 19,481 ER visits from 2019. The claim does not hold.

II. The Access Override: What the Data Shows

Ambulance arrivals wait less than walk-in arrivals at every triage level without exception.

Triage is the clinical ranking system ERs use to prioritize patients by severity. Level 1 is a life-threatening emergency. Level 5 is a minor condition that could be treated elsewhere. The system is designed so that clinical severity — not anything else — determines who gets seen first.

The data shows that arrival method determines wait time independent of that system. At every one of the five triage levels, ambulance arrivals are seen faster than walk-in arrivals classified at the same severity. The median gap runs between five and seven minutes. It does not disappear at the most severe end. At triage level 1 —

where both groups have been classified by a clinician as requiring immediate intervention — ambulance arrivals wait a median of 6 minutes. Walk-in arrivals wait a median of 13 minutes.

That is the access override. Ambulance arrival routes patients ahead of triage priority. The routing mechanism, not the clinical severity, is determining how quickly you are seen.

III. Every Challenge Tested. None Survived.

A finding this consistent invites challenge. The audit was designed to test every reasonable alternative explanation before drawing a conclusion.

The Triage Bias Critique

Triage levels are assigned by clinicians who know how a patient arrived. A critic could argue that clinicians unconsciously rate ambulance patients as more severe because of the arrival method itself — not the actual condition. To test this, the analysis used a second severity measure that clinicians cannot influence: the patient's own reported pain score, recorded at triage before any clinical assessment. The access override holds at every pain level from 0 to 10. At maximum reported pain, ambulance arrivals wait 8 minutes. Walk-in arrivals reporting the same pain wait 15 minutes. Two independent severity measures confirm the same signal. The triage bias critique does not survive.

The Boarding Congestion Hypothesis

Boarding occurs when admitted patients remain in the ER waiting for an inpatient bed. A boarded patient occupies resources and contributes to congestion. The hypothesis was that ambulance arrivals — who are more likely to be admitted — generate the congestion that slows walk-in wait times. If true, boarding would be the real mechanism.

Ambulance arrivals are boarded at a rate of 20.67%. Walk-in arrivals are boarded at 6.19% — less than a third of the ambulance rate. Ambulance arrivals generate significantly more boarding burden and still wait less. The boarding hypothesis does not explain the gap. It makes the finding stronger: the access override effect survives a structural congestion disadvantage that should, under this hypothesis, have produced the opposite result.

The Regional Anomaly Question

If the finding were driven by one region's routing practices, it would not be a national claim. The analysis tested all four US Census regions. The gap holds in every one — Northeast (13 minutes), Midwest (7 minutes), South (3 minutes), West (3 minutes). No region eliminates the advantage. The access override is a national pattern.

The Operational Conditions Question

ERs operate differently at 2am than at 2pm, and differently on Mondays than on Saturdays. If the gap were driven by specific operational conditions rather than arrival routing, it would appear in some windows and disappear in others. It does not. The gap holds across all four time-of-day bands and all seven days of the week. Monday shows the widest gap — consistent with elevated walk-in volume from patients who delayed care over the weekend — but the gap does not disappear on any day.

IV. Why the Mean Almost Hid the Finding

The correct statistic for this dataset is the median, not the mean. ER wait times contain extreme outliers — some patients wait over twenty hours. The mean is pulled upward by those cases in ways that obscure the structural pattern. The median is not.

The Northeast region shows this most clearly. Ambulance arrivals in the Northeast have a mean wait time of 56.0 minutes. Walk-in arrivals have a mean of 56.3 minutes. A viewer relying on means would conclude the access override disappears entirely in the Northeast. The medians show 13 minutes versus 26 minutes — a 13-minute gap that the mean hides completely. The same phenomenon appears on Wednesdays, where ambulance and walk-in means are nearly identical while the medians show a 6-minute gap. The mean anomaly is not a regional artifact. It is a feature of this data that appears wherever outliers are concentrated, and it is precisely why median is the lead statistic throughout this analysis.

V. What This Means

The folk wisdom is correct. Going to the ER by ambulance gets you seen faster — consistently, at every severity level, in every region of the country, at every time of day, on every day of the week.

The official messaging telling the public otherwise is not supported by 2019 federal data.

This audit does not establish why the gap exists. The dataset does not record routing pathways — it does not show whether ambulance patients are taken directly to treatment areas while walk-in patients enter the waiting room queue. The access override theory is consistent with how ERs are physically and operationally structured, but the mechanism requires facility-level operational data to confirm. What is confirmed here is the pattern: arrival method determines wait time, independent of clinical severity, across every dimension tested.

A publicly held belief was tested against federal data. The belief was confirmed. The official messaging telling people otherwise is not supported by the data.

VI. The Numbers

- 19,481 total ED visit records — 2019 CDC NHAMCS nationally representative sample
- Raw dataset originally distributed by CDC in Stata format (.dta) — converted to CSV using Python and pandas prior to analysis
- 11,512 records in the clean analytical universe — 59.09% of total after data quality filtering
- 5–7 minutes — consistent median wait time advantage for ambulance arrivals across all five triage levels
- 5 of 5 triage levels confirm the access override
- 4 of 4 US Census regions confirm the access override
- 4 of 4 time-of-day bands confirm the access override
- 7 of 7 days of the week confirm the access override
- 11 of 11 pain score levels (0–10) confirm the access override
- 20.67% ambulance boarding rate vs. 6.19% walk-in boarding rate — alternative explanation ruled out

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Analysis conducted June 10–11, 2026. Dataset: CDC NHAMCS 2019 Emergency Department Public Use File. Full methodology documented in 04_EDAOA_Master_StepByStep.sql. Cross-platform validation documented in 05_EDAOA_Python_Cross_Platform.ipynb. GitHub: ed-access-override-audit.